

Assignment-1

Question-1:

Load the contents of memory location 2200_{16} and 2201_{16} into registers. Add these registers and store result in memory locations 2202_{16} and 2203_{16} .

Solution:

Algorithm:

- start
- Load address 2200 to HL register
- Load contents of 2200_{16} to Accumulator.
- Increment address in HL register to 2201_{16}
- Load contents of 2201_{16} to B register.
- Add contents of B to accumulator and store carry.
- Store value in accumulator (sum) to 2202_{16}
- Reset the accumulator to $00H$
- Add the carry bit to accumulator.
- Store contents of accumulator (carry) to 2203_{16}
- End

Code:

<u>Opcode</u>	<u>Hex code</u>	<u>Description</u>
LXI H, 2200H	21 00 22	Loads address to HL
MOV A, M	7E	Moves contents of HL to A
INX H	23	Increment HL
MOV B, M	46	Move content of HL to B
ADC B	88	Add B to A with carry
STA 2202H	32 02 22	Store A to address
MOV A, 00H	3E 00	Make A = 0
ADC A	8F	Add carry bit to A.
STA 2203H	32 03 22	Store A to address
HLT	76	End of program. (Halt)

Question-2 :

Find sum of N numbers starting from E100H, value of N is stored in E200H. Store the result in locations E300H and E301H.

Solution:Algorithm

→ Start

→ Load N to Accumulator.

Move value of accumulator to B register.

$A \leftarrow 00H$

$C \leftarrow 00H$

$1H \leftarrow E100H$

Label loop:

$A = A + M$ (Adds data from HL register to A)

~~$1H \leftarrow 1H + 1$~~

if (! carry)

goto next

$[C] = [C] + 1$

Label next:

$B = B - 1$

if (! zero)

goto loop

$E300H \leftarrow A$

$A \leftarrow C$

$E301H \leftarrow A$

End

Code:

<u>Opcode</u>	<u>Hexcode</u>	<u>Description</u>
LDA E200H	3A 00 E2	Content of E200 to A
MOV B, A	47	Contents of A to B
MVI A, 00H	3E 00	A = 0
MVI C, 00H	0E 00	C = 0
LXI H, E100H	21 00 E1	Load address E100 to HL
loop: ADD M	86	Add contents of HL to A
INX H	23	increment address in HL
JNC next	D2 11 E0	Jump to next if no carry
INR C	0C	increment content in C.
next: DCR B	05	decrement content of B.
JNZ loop	C2 06 E0	Jump to loop if not zero
STA E300H	32 00 E3	store A to E300
MOV A, C	79	contents of C to A.
STA E301H	32 01 E3	store A to E301
HLT	76	End

Note! Addresses used in question are not identical to what was given in assignment. This is done as the given addresses were not present in the machine provided in lab.

Question-3

Find sum of least significant 4 bits and most significant 4 bits of a byte stored in memory location E100H. Store the result in E300H.

Solution:

Algorithm:

Start

$D \leftarrow [E300H]$

$A \leftarrow [E100H]$

$B \leftarrow A$

$A = (A \& 0FH)$

$C \leftarrow A$

$A \leftarrow B$

$A = (A \& F0H)$

Rotate A to left by 4 bits

$A = A + C$

$[D] \leftarrow A$ (store value in A to E300H)

End

Code:

Instructions start from E000H

Data taken from E100H

Result stored in E300H

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<u>opcode</u>	<u>Hexcode</u>	<u>Description</u>
LXI D, E300H	11 00 E3	Load immediate to D
LDA E100H	3A 00 E1	Load Direct to A
MOV B, A	47	move contents of A to B
ANI 0F H	E6 0F	bitwise and of 0F and A.
MOV C, A	4F	move contents of A to C
MOV A, B	78	move contents of B to A.
ANI F0 H	E6 F0	bitwise and of F0 and A
RLC	07	Rotate Left
RLC	07	
RLC	07	
RLC	07	
ADD C	81	Add contents of C to A
STAX D	12	store value in A to address in D
HLT	76	Halt.

Observations:

$$[E100] = (AB)_{16}$$

After execution:

$$[E300] = (15)_{16}$$

$$((A)_{16} + (B)_{16} = (15)_{16})$$

Hence the experiment was a success.

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Question 4:

Write a program to count the 0's and 1's of a byte stored E100H. Store the result in E300H and E301H respectively.

Solution:Algorithm:

Start

H $\leftarrow$ E301H

A $\leftarrow$ [E100H]

B $\leftarrow$ A

C $\leftarrow$ 08H

label x: Rotate accumulator bits to right

if (!carry)

goto y

D $\leftarrow$ D+1

label y: C $\leftarrow$ C-1

if (!zero)

goto x

[HL] $\leftarrow$ D

C $\leftarrow$ 08H

A $\leftarrow$ C

A $\leftarrow$ A-D

H $\leftarrow$ E300H

[HL] $\leftarrow$ A

End

Code:

<u>opcode</u>	<u>Hex code</u>	<u>Description</u>
LXI H, E301H	21 01 E3	Load Immediate HL
LDA E100H	3A 00 E1	Load Direct Accumulator
MOV B, A	47	move content of A to B
MVI C, 08H	0E 08	move immediate C
X: RAR	1F	Rotate Accumulator Right
JNC Y	D2 00 E0	Jump not carry to y
INR D	14	increment value of D.
Y: DCR C	0D	decrement value of C.
JNZ X	C2 09 E0	Jump to x if not zero.
MOV M, D	72	move content of D to memory
MVI C, 08H	0E 08	move immediate C
MOV A, C	7D	move content of C to A.
SUB D	92	Subtract D from A.
LXI H, E300H	21 00 E3	Load immediate HL
MOV M, A	77	move content of A to memory
HLT	76	End.

Observations:

Data: $(AB)_{16} = (1010\ 1011)_2$

$\therefore [E100] = (1010\ 1011)_2$

After execution: $[E300] = 3 = \text{no of zeros}$
 $[E301] = 5 = \text{no of ones}$

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Write a program to sum two 16-bit binary numbers.

Data stored in:

E100H (LSB)	, E101H (MSB)	→ 1st 16 bit number
E102H	, E103H	→ 2nd 16 bit number

Algorithm:

$$L \leftarrow [E100M]$$
$$14 \leftarrow [10111]$$

DE $\leftarrow$ HL

$$L \leftarrow [E102H]$$
$$14 \leftarrow [E103H]$$

C ← 0014

$HL \leftarrow HL + DE$ (Register pair contents are added)

if (!carry)

goto loop

$C \leftarrow C + 1$

label loop: A ← C

[E302H] ← A

$$[E30114] \leftarrow 14$$
$$[E=300\text{H}] \leftarrow L$$

End

Code:

<u>opcode</u>	<u>Hexcode</u>	<u>Description</u>
LHLD E100H	2A 00 E1	Load HL direct
XCHG	EB	Exchange HL and DE
LHLD E102H	2A 02 E1	Load HL direct
MVI C, 00H	0E 00	move 00H to C
DAD D	19	Double add HL and DE
JNC loop	D2 0E E0	Jump to loop if no carry
INR C	0C	increment C
loop: MOV A, C	79	move content of C to A.
STA E302H	32 02 E3	Store content of A to address
SHLD E300H	22 00 E3	Store value of HL register pair
HLT	76	End

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